

Introduction

An achievement test measures how an individual has learned over time and what the individual has learned by analysing his present performance. It also measures how a person understands and masters a particular knowledge area at the present time.

Achievement tests are also a type of psychological tests that measure your ability to comprehend a specific topic, for example, mathematics (Blog, 2013).

Purpose of Achievement Test

The purpose of achievement testing is to measure some aspect of the intellectual competence of human beings: what a person has learned to know or to do. Teachers use achievement tests to measure the attainments of their students. Employers use achievement tests to measure the competence of prospective employees. Professional associations use achievement tests to exclude unqualified applicants from the practice of the profession. In any circumstances where it is necessary or useful to distinguish persons of higher from those of lower competence or attainments, achievement testing is likely to occur (Ebel, 1965).

Construction of Achievement Test

For constructing achievement test in any school examinee. Following two questions are gives a way to construct the achievement test.

- a. What to measure?
- b. How to measure?
- a. What to measure involves certain objectives in view that enables to bring accepted changes in the student's behaviour when we teach.

- b. How to measure involves how to construct an instrument that measures the abilities of students (Scott, 2010).

Standardization

Standardization is the process of high qualities and deep understanding of examinee matter. Standardization test specifies the directions for administrating and scoring the test on different groups of individuals like whom that the test is designed. Standard content develops the procedure for identical test on individuals in different places and at different times (Scott, 2010).

According to Kaplan and Saccuzzo (2009), following are the steps for constructing an achievement test.

1. Planning the Test

- Designing the Test
- Preparation of Blueprint

2. Preparing Preliminary Draft

- Item Writing
- Item Editing
- Pre Try Out

3. The Try Out

4. Item Analysis

5. Preparing the Final Draft

6. Establishment of

- Reliability
- Validity
- Norms

Description of all above steps is described below briefly.

Planning the Test

Planning of achievement test will be carried out with the help of two steps.

Designing the Test. Designing is most important step in the building test.

Designer should be careful about planning and making test successful. He should keep this in mind that how and what types of questions are used. He has to decide weightage for different objectives and units in the course. Following points helps designer to design test in systematic way.

- Identification of objectives
- Measurement of content
- Allotment of time
- Allotment of marks
- Include multiple choice question
- Emphasis on each examinee and area of question

Preparation of Blueprint. After designing preparation of blueprint is the last stage of the planning of test. Here test constructor put various type of question in blueprint and allots them marks depending on the time. The tester writes down his decisions in the form of a blueprint. With the completion of blueprint the remaining decisions of the design will become the basis for writing the items.

Preliminary draft

Preliminary draft is prepared by following three stages.

Item Writing. It is important step in preliminary draft and here blue print is used as a guide writing this draft. Test conductor should have following points in mind when he writes items for preliminary draft.

- Each item contains single idea.
- Questions should be clear.

- Simple and easy to understand.
- Double barrelled question should be avoided.
- Arrangement of items should be from simple to complex.
- Subjective question should also be avoided.

Item Editing. Then the item should be edited and reviewed by language teacher and also by experts of measurement. Language teacher will check the errors in language and defect in words. By removing defects it is submitting to experts. Experts of measurement measure the level of achievement of items.

Pre Try Out. The draft is modified and remove the shortcomings of preliminary draft. In this stage constructor is confident for the usability of test.

The Try Out

To ensure the proper operating of test items and to remove shortcoming, it is essential to have try out of items. This helps in predict how students will work in actual practices. At the try out stage, the test should be so timed that nearly 90% of the sample should be able to attempt all the items of the test. The test sheets along with answer sheets are collected and answer sheets are used for scoring with the help of scoring keys. Keep following points in mind while testing in sample.

- Proper sitting arrangements.
- Time of administrating the test.
- Total time required for test.
- Proper motivation with the pupil.

Item Analysis

In this stage test constructor examining the responses of respondents in the sample to each test item. It can also be defined as it is a statistical procedure by which the

appropriate items are selected for the final draft and poor items are rejected. Item analysis is an analysis of responses made to 'teacher made tests' by the pupil in the class.

Final Draft

In the final draft questions should be arranged in such way that easy, average and difficult and again starts from easy and so on. By doing this student at least go through whole test because he know that some of remaining questions are easy. When the test is arranged properly with the help of principles, a clear identity of test is appeared.

Establishment of Reliability, Validity and Norms

With the selection of good items final draft has been prepared and the final step is the establishment of reliability, validity and norms. These test items administered to a larger sample.

Reliability. The reliability of a test measures whatever it measure. There are many methods for computing the reliability of a test and the most appropriate method for computing reliability of achievement test is Split-half method.

Validity. It is very important aspect of test and it can be determined as the degree which is capable of measuring achievements and it is design to do so. Valid test is highly reliable. Validity is measured by four methods.

- Face validity
- Content validity
- Concurrent validity
- Predictive validity

Norms. It is average score of sample population. These are the level obtained by a particular group of persons on a test. There are many types of norms like age norms, grade norms, percentile norms and standard scores.

Method

Participants

50 students of Grade 9 science section were taken from Air Foundation School PWD, Islamabad. There were 23 girls and 27 boys.

Procedure

Test Construction

A self-made achievement test of 50 multiple choice items was prepared for the subject of biology. The items were prepared based on the following cognitive domains

1. Knowledge
2. Comprehension
3. Application
4. Analyses

Knowledge. Involves the recall of specifics and universals, the recall of methods and processes, or the recall of a pattern, structure, or setting.

Comprehension. Refers to a type of understanding or apprehension such that the individual knows what is being communicated and can make use of the material or idea being communicated without necessarily relating it to other material or seeing its fullest implications.

Application. Refers to the “use of abstractions in particular and concrete situations.

Analysis. Represents the “breakdown of a communication into its constituent elements or parts such that the relative hierarchy of ideas is made clear and/or the relations between ideas expressed are made explicit.

Specification Chart

Objectives Form of questions (MCQs)	Knowledge (k)	Understanding (u)	Application (app)	Analysis (analyz)	Grand total
Topic 1 Introduction to biology	10	5	1	1	17
Topic 2 Solving a biological	6	5	5	1	17
Topic 3 Biodiversity	10	4	0	2	16
Total marks	26	14	6	4	50

Test Administration

Students were given instructions to fill in the demographic details and encircle the correct option, there will be no negative marking and each MCQ carry equal marks. 1 hour was allotted for the test completion.

Item Analysis

Item analysis is a process which examines student responses to individual test items (questions) in order to assess the quality of those items and of the test as a whole. Item analysis is especially valuable in improving items which will be used again in later

tests, but it can also be used to eliminate ambiguous or misleading items in a single test administration (Lehmann, 2020).

Item Difficulty. For items with one correct alternative worth a single point, the item difficulty is simply the percentage of students who answer an item correctly. In this case, it is also equal to the item mean. The item difficulty index ranges from 0 to 1; the higher the value, the easier the question. Excel sheet was used to calculate item difficulty index.

Item Difficulty Range

Range	Comments
0-0.20	Very Difficult
0.21-0.40	Difficult
0.41-0.60	Moderate
0.61-0.80	Easy
0.81-1	Very Easy

Item Difficulty Index

Serial no.	Qs no.	Item difficulty	Comments
1	Q1	0.88	Very Easy
2	Q2	0.58	Moderate
3	Q3	0.84	Very Easy
4	Q4	0.42	Moderate
5	Q5	0.48	Moderate
6	Q6	0.78	Easy
7	Q7	0.26	Difficult
8	Q8	0.60	Easy
9	Q9	0.08	Very Difficult
10	Q10	0.36	Difficult
11	Q11	0.68	Easy
12	Q12	0.62	Easy
13	Q13	0.58	Moderate
14	Q14	0.28	Difficult
15	Q15	0.60	Moderate

16	Q16	0.01	
17	Q17	0.58	Very Difficult
18	Q18	0.40	Moderate
19	Q19	0.84	Difficult
20	Q20	0.80	Very Easy
21	Q21	0.58	Very Easy
22	Q22	0.38	Moderate
23	Q23	0.14	Difficult
24	Q24	0.46	Very Difficult
25	Q25	0.78	Moderate
26	Q26	0.62	Easy
27	Q27	0.64	Easy
28	Q28	0.38	Easy
29	Q29	0.30	Difficult
30	Q30	0.46	Difficult
31	Q31	0.68	Moderate
32	Q32	0.46	Easy
33	Q33	0.38	Moderate
34	Q34	0.16	Difficult
35	Q35	0.22	Very Difficult
36	Q36	0.42	Difficult
37	Q37	0.60	Moderate
38	Q38	0.56	Moderate
39	Q39	0.48	Moderate
40	Q40	0.38	Moderate
41	Q41	0.58	Difficult
42	Q42	0.74	Moderate
43	Q43	0.74	Easy
44	Q44	0.68	Easy
45	Q45	0.34	Easy
46	Q46	0.48	Difficult
47	Q47	0.58	Moderate
48	Q48	0.20	Moderate
49	Q49	0.20	Very Difficult
50	Q50	0.18	Very Difficult

The above bold items were discarded because they were too easy and too difficult.

Item Discrimination

Item discrimination refers to the ability of an item to differentiate among students on the basis of how well they know the material being tested. Excel sheet was used to calculate discrimination index.

Item Discrimination Range

Range	Comments
0-0.24	Bad
0.25-0.39	Good
0.40-1	Excellent

Item Discrimination Index

Serial no.	Q no.	Item discrimination	Comments
1	Q1	0.22	Bad
2	Q2	0.05	Bad
3	Q3	0.36	Good
4	Q4	0.60	Excellent
5	Q5	0.51	Excellent
6	Q6	0.36	Good
7	Q7	0.53	Excellent
8	Q8	0.10	Bad
9	Q9	-0.08	Bad
10	Q10	0.66	Excellent
11	Q11	-0.25	Bad
12	Q12	0.31	Good
13	Q13	0.17	Bad
14	Q14	0.19	Bad
15	Q15	0.44	Excellent
16	Q16	-0.29	Bad
17	Q17	0.30	Good
18	Q18	0.45	Excellent
19	Q19	0.50	Excellent
20	Q20	0.15	Bad
21	Q21	-0.17	Bad
22	Q22	0.80	Excellent
23	Q23	0.12	Bad
24	Q24	0.04	Bad
25	Q25	0.16	Bad
26	Q26	0.50	Excellent
27	Q27	0.44	Excellent
28	Q28	0.04	Bad
29	Q29	0.46	Excellent
30	Q30	0.52	Excellent
31	Q31	0.37	Good

32	Q32	0.52	Excellent
33	Q33	0.59	Excellent
34	Q34	0.27	Good
35	Q35	-0.09	Bad
36	Q36	0.31	Good
37	Q37	0.44	Excellent
38	Q38	0.50	Excellent
39	Q39	0.51	Excellent
40	Q40	0.79	Excellent
41	Q41	0.58	Excellent
42	Q42	0.37	Good
43	Q43	0.58	Excellent
44	Q44	0.24	Bad
45	Q45	-0.02	Bad
46	Q46	0.10	Bad
47	Q47	0.52	Excellent
48	Q48	0.46	Excellent
49	Q49	0.26	Good
50	Q50	0.12	Bad

The above bold items were discarded due to bad discrimination index.